

Stability-derived upper limits on the true masses of non-transiting radial-velocity planets: a homogeneous N -body survey

Shaurya Ranjan^{1*}

¹*Independent Researcher*

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ABSTRACT

The radial-velocity method measures a planet’s minimum mass, $m \sin i$, but leaves the orbital inclination i , and therefore the true mass, unconstrained. In a multi-planet system, long-term dynamical stability supplies the missing bound. Because each planet’s true mass scales as $1/\sin i$, a system viewed near face-on contains heavier planets that perturb one another more strongly, and above some inclination the configuration no longer survives for the age of the system. The surviving range of inclinations therefore sets an upper limit on each planet’s true mass. This argument has been applied to individual systems with a variety of methods, but no uniform survey exists. We carry out a homogeneous N -body stability survey of 56 multi-planet radial-velocity systems drawn from the NASA Exoplanet Archive. For each system we sweep the line-of-sight inclination over a fixed grid and sample all orbital parameters within their published uncertainties. We validate the pipeline on GJ 876, recovering the published stability constraint ($i > 25^\circ$, against the literature value $i \gtrsim 20^\circ$). Under an isotropic prior on orientation we derive 95 per cent credible upper limits on the true mass of 123 planets in 53 systems. For 61 planets in 23 systems the limit falls below $3.0 m \sin i$, tighter than geometry alone provides (median $2.68 m \sin i$), and 25 giant planets are confined below the deuterium-burning limit on dynamical grounds. We also document three ways in which tabulated catalog parameters can be dynamically inconsistent, and describe how the survey guards against each.

Key words: celestial mechanics – planets and satellites: dynamical evolution and stability – techniques: radial velocities – methods: numerical – catalogues

1 INTRODUCTION

The radial-velocity method detects a planet through the reflex motion it induces in its host star along the line of sight. This motion fixes the product $m \sin i$, where m is the planet mass and i the orbital inclination, but not m itself. For a randomly oriented orbit the correction from $m \sin i$ to m is usually small: the median value of $1/\sin i$ is 1.15, and 95 per cent of orientations give a true mass below $3.2 m \sin i$. For any individual planet, however, $m \sin i$ is a lower bound rather than a measurement. This one-sided uncertainty propagates into quantities built on radial-velocity masses. Mass functions and occurrence rates are biased low; the boundary between planets and brown dwarfs is ambiguous for any giant whose true mass could exceed $13 M_{\text{Jup}}$; and inferences about the formation and composition of individual systems often treat a lower limit as if it were the mass.

In a system of two or more planets, dynamics provides the upper bound that the radial-velocity data lack. Consider a coplanar system tilted as a rigid body towards a face-on orientation. Its internal dynamics are unchanged, but every $m \sin i$

mass increases as $1/\sin i$. Heavier planets exert stronger mutual perturbations, and for a sufficiently face-on orientation the system becomes unstable on a time-scale far shorter than its age. Such configurations cannot describe a system that is observed to exist today, so the requirement of long-term survival places a ceiling on each planet’s true mass. This argument is well established for individual systems. GJ 876 is the canonical example; stability has also been used to constrain 47 UMa (Goździewski 2002), HD 141399 (Vogt et al. 2014), and Barnard’s star (Basant et al. 2025), and Tamayo et al. (2021) developed it into a general method for stability-constrained characterization, which they applied to Kepler-23. What has been missing is a uniform treatment. Laskar & Petit (2017) classified 131 systems by AMD stability but did not convert stability into per-planet mass limits; Volpi et al. (2019) used first-order secular theory to bound the mutual inclinations of ten radial-velocity systems; and Qin et al. (2025) propose recovering inclinations from von Zeipel–Lidov–Kozai cycles in compact binaries. The remaining literature consists of single-system analyses that adopt different integrators, instability criteria, integration times, and statistical definitions, so their mass limits cannot be compared and no population-level statement can be drawn from them.

* E-mail: shauryaranjan911@gmail.com

This paper presents a homogeneous survey that applies one pipeline, one instability criterion, and one statistical definition to every qualifying multi-planet radial-velocity system in the NASA Exoplanet Archive. Orbital parameters are sampled within their published uncertainties, and the pipeline is validated against systems whose stability constraints are already known. We report 95 per cent credible upper limits on true mass under an isotropic orientation prior, a set of systems flagged as parameter-inconsistent, and three classes of catalog-parameter pitfall that the dynamical treatment exposes (Section 3.4).

2 SAMPLE SELECTION

We draw candidate systems from the composite parameters table (`pscompars`) of the NASA Exoplanet Archive, accessed 2026 July 9–13. A system qualifies if it contains at least two planets with measured periods and masses, has a measured stellar mass, and contains at least one non-transiting planet whose tabulated mass is a minimum mass ($m \sin i$); these are the planets the method can constrain. Because mutual perturbations weaken rapidly with orbital separation, we also require at least one adjacent planet pair with a period ratio below 6. These criteria yield 148 systems.

We rank each system by an analytic estimate of the mass factor at which it should become unstable. For the most fragile adjacent pair, with masses scaled by a common factor f and separation measured in eccentricity-reduced mutual Hill radii, we find the value f_{crit} at which the pair reaches the two-planet overlap boundary $\Delta = 2\sqrt{3}$ (Gladman 1993). Systems of more than two planets destabilize at wider spacings, so f_{crit} overestimates the true ceiling and is therefore a conservative selection criterion. We survey the band $1.05 < f_{\text{crit}} < 60$ with maximum eccentricity $e_{\text{max}} \leq 0.75$ and inner-planet period $P_{\text{inner}} \geq 1.5$ d, which contains 56 systems. Systems with $f_{\text{crit}} \approx 1$ are excluded: this group is dominated by resonant and near-resonant pairs whose survival depends on the relative orbital phases, information that the composite catalog does not supply reliably (Sections 3.4 and 6). Table 1 lists the disposition of all 148 systems: 53 surveyed, 3 flagged as parameter-inconsistent, and 92 excluded. The excluded systems are resonance-protected (42), too weakly interacting for stability to constrain the mass at any allowed value (48), or have an inner period below 1.5 d (9), with some systems in more than one category.

3 METHODS

3.1 Configuration sampling

Tilting a coplanar system rigidly with respect to the line of sight changes the inferred masses but not the internal dynamics: every $m \sin i$ mass scales as $1/\sin i$ while all mutual angles are preserved. We therefore treat inclination as the single swept parameter. We sample i on a 17-point grid from 90° (edge-on, masses at their minimum values) to 5° (masses at 11.5 times the minimum). Planets with a true mass known from a transit or from astrometry are held fixed. At each grid point we integrate 20 independent realizations. In each realization we draw the stellar mass and every planet’s mass, period, eccentricity, and argument of periastron from truncated

normal distributions defined by the symmetrized 1σ catalog uncertainties; where an uncertainty is not tabulated we use the point value, and eccentricities are truncated to $[0, 0.95]$. Angles the catalog does not report are drawn uniformly. Orbital phases are randomized for systems whose parameters are assembled from multiple references, and taken from the fitted time of periastron only for systems with a single self-consistent reference, for the reasons given in Section 3.4.

3.2 Integration and instability criterion

We integrate each realization with the WHFast symplectic integrator (Rein & Tamayo 2015) in the REBOUND package (Rein & Liu 2012). The time-step is $dt = \min_p P_p (1 - e_p)^{3/2} / 20$, one-twentieth of the shortest perihelion passage time in the system, which keeps the fastest pericentre well resolved. A realization is recorded as unstable if any two bodies approach within the sum of their Hill radii (each body carries its Hill radius as a collision radius) or if any body moves beyond 20 times the widest semi-major axis. Each realization is integrated for a fixed number of inner-planet orbital periods rather than a fixed physical time, so the computational cost per system does not depend on the absolute scale of the system. We use two depths, 10^5 and 10^6 inner orbits; Section 4.3 compares them.

We test the sensitivity of the limits to these choices on a representative subset of six systems that span the survey’s range of architectures. Halving the time-step changes the 95 per cent credible limits by at most 0.03 in the mass factor. Reducing the collision radius to 0.5 and 0.25 times the Hill radius changes them by at most 0.08. Replacing exact coplanarity with mutual inclinations drawn from Rayleigh distributions of width $\sigma = 2^\circ$ and 5° changes them by at most 0.4. None of these shifts exceeds the grid spacing in inclination.

3.3 Statistical definitions

For each system we construct a survival curve $S(i)$, the fraction of the 20 realizations at inclination i that remain stable to t_{max} . We define the constraint relative to the edge-on value $S(90^\circ)$, because a system whose published parameters are already marginal can fail even at edge-on orientations. A system with $S(90^\circ) < 0.5$ is reported as a parameter-consistency flag; its result describes the published parameters, not a mass limit. For the remaining systems we report a 95 per cent credible upper limit on the mass factor under an isotropic prior on orientation (uniform in $\cos i$), taking the posterior density proportional to $S(i)$; the quoted limit is the mass factor below which 95 per cent of the surviving posterior lies. With no dynamical information this limit equals 3.20 for every planet, so values below 3.20 measure the information the survey adds. For comparison with earlier single-system work we also tabulate a threshold ceiling, defined as the smallest mass factor at which S falls below half its edge-on value.

3.4 Catalog-parameter pitfalls guarded against

During development we encountered three recurring ways in which tabulated catalog parameters can be dynamically inconsistent. The survey guards against each.

(i) *Eccentricity point estimates incompatible with survival.* Compact systems are often fit with small, poorly constrained eccentricities whose point values nonetheless destabilize the system when integrated directly. For Barnard’s star the tabulated values $e = 0.03\text{--}0.08$ destroy the system within a few thousand years, whereas eccentricities drawn from within the quoted uncertainties do not; the same conclusion was reached independently by [Basant et al. \(2025\)](#), whose fits favour $e < 0.02$. We therefore sample eccentricities within their uncertainties and never integrate the point values alone (Section 3.1).

(ii) *Composite parameters from incompatible fits.* The `pscomppars` table assembles each parameter from the best available source, so a single system can combine values from different fits. For resonant systems this destroys the phase relationships on which stability depends. We randomize the orbital phases of such systems and take fitted phases only from a single self-consistent reference (Section 3.1).

(iii) *Fit epochs recorded as periastron times.* For some references the tabulated time of periastron is the epoch of the fit, which is identical for every planet and, if read literally, places all planets at periastron simultaneously (GJ 876; [Rivera et al. 2010](#)). We reject a system’s periastron times when they are identical across its planets before using any phase information.

4 VALIDATION

We validate the pipeline on two systems chosen to bracket the two dynamical regimes it encounters: GJ 876, whose survival depends on the relative phases of a mean-motion resonance, and HD 45364, which is stable independently of its phases. A third, external check appears in Section 5.1, where the survey’s mass ceilings for Barnard’s star are compared with the independent stability masses of [Byrne et al. \(2026\)](#).

4.1 GJ 876: the phase-protected anchor

GJ 876 is the standard test case for stability-constrained characterization. The Archive’s default solution is the coplanar fit of [Rivera et al. \(2010\)](#) at $i = 59^\circ$, which lists true masses and places the 4:2:1 Laplace resonance at the centre of its libration. We reconstruct the equivalent minimum-mass system by scaling the fitted masses by $\sin 59^\circ$ and fix the mean anomalies to the values in their Table 3. For this reference the Archive’s time-of-periastron column holds the epoch of the fit rather than a per-planet periastron passage (pitfall 3 of Section 3.4), so the phases must be taken from the paper. We then sweep a 13-point inclination grid with the phases held fixed, using 5 realizations per grid point and 5000 yr ($\approx 10^6$ inner orbits) per realization.

The test has three requirements (Fig. 1). First, the published configuration, which enters the sweep at mass factor $1/\sin 59^\circ = 1.17$, must survive; it does. Secondly, the recovered ceiling must match the published constraint: survival is zero for all grid inclinations at or below 25° and non-zero above, giving $i > 25^\circ$ ($m < 2.37 m \sin i$), consistent with and slightly more conservative than the literature value $i \gtrsim 20^\circ$. Thirdly, the same masses with randomized phases must not remain stable, since otherwise resonant protection would play

no role; survival falls to 12 per cent. The pipeline thus reproduces both the resonant protection and the known inclination limit.

4.2 HD 45364: the phase-agnostic anchor

HD 45364 tests the opposite regime. Its two giant planets lie in a 3:2 commensurability with low eccentricities ([Correia et al. 2009](#)), and the [Li et al. \(2022\)](#) solution is stable without help from resonant phases. Initialized with the fitted phases, the published configuration survives the full 10^5 yr integration, and randomizing the phases leaves survival unchanged at 100 per cent. The survival curve is correspondingly smooth rather than sharpened by resonance: complete survival for $i \geq 45^\circ$ falling to zero by $i \approx 8.5^\circ$ (17-point grid, 10 realizations per point). Under the isotropic prior this gives a 95 per cent credible limit of $m < 2.29 m \sin i$. The two anchors bracket the survey population, one system constrained by its phases and one independent of them, and the pipeline reproduces both without any per-system adjustment.

4.3 Depth convergence

Because we integrate for a fixed number of orbits rather than to a physical age, we must show that the reported limits reflect the dynamics and not the integration length. We therefore run every system at two depths, 10^5 and 10^6 inner orbits, using identical parameter draws (the random seed is fixed by the system name), and compare the threshold ceilings. A longer integration can only reveal additional instability, so the ceilings should move in one direction. They do: among the 53 unflagged systems, 9 ceilings tighten between the two depths, 5 are unchanged, none loosen, and 7 systems that show no ceiling at 10^5 orbits acquire one at 10^6 (Fig. 3). This is the third successive generation of the survey to show strictly one-directional behaviour, which is what allows us to describe the reported limits as conservative: a longer integration can only strengthen them (Section 6).

As a stronger test of the same property, we re-ran at 10^7 inner orbits the 16 systems whose ceilings had moved between 10^5 and 10^6 orbits. Eight ceilings tighten further and eight are unchanged; none loosen, so the one-directional behaviour persists at a tenfold-longer integration.

5 RESULTS

5.1 The catalogue

Of the 56 systems in the survey band, 53 pass the edge-on consistency test and yield 95 per cent credible mass limits for 123 non-transiting planets (Table 2; Fig. 2). For 61 planets in 23 systems the limit falls below $3.0 m \sin i$, inside the value of 3.20 that geometry alone guarantees, with a median of $2.68 m \sin i$ among them. Twenty-five giant planets whose face-on orientations would place them above the deuterium-burning limit are instead confined below it at 95 per cent credibility.

Several individual results illustrate the range. The four sub-Earth planets of Barnard’s star are limited to $2.18 m \sin i$ ($0.42\text{--}0.73 M_\oplus$), in agreement with the independent stability masses of $0.19\text{--}0.84 M_\oplus$ reported by [Byrne et al. \(2026\)](#).

The three giant planets of 47 UMa are limited to $2.35 m \sin i$ and the four giant planets of HD 141399 to $2.71 m \sin i$. K2-18 c, the non-transiting companion of the habitable-zone sub-Neptune K2-18 b, is limited to $2.87 m \sin i \approx 21.6 M_{\oplus}$, which excludes a giant-planet mass. For K2-18 c the isotropic prior is conservative: because K2-18 b transits, only a small mutual inclination is needed for c to avoid transiting, so near-edge-on orientations are favoured a priori (Cloutier et al. 2017).

The tightest limit in the catalogue shows that a small mass factor and a planetary-mass classification are separate properties. The pair around TYC 1422-614-1 is limited to $2.01 m \sin i$, the strongest relative constraint in the survey, but the minimum mass of the outer planet is already $10 M_{\text{Jup}}$, so even this limit permits a mass of $\approx 20 M_{\text{Jup}}$, above the deuterium-burning limit.

5.2 Population statement

Fig. 4 shows the cumulative distribution of the 123 credible limits. Half the sample (61 planets) is constrained below the geometric value of 3.20, and the tightest fifth below $2.6 m \sin i$. No limit exceeds 3.20 by more than the discretization of the inclination grid; the single nominal exception, TOI-1062 c at 3.22, is an unconstrained posterior evaluated on the 17-point grid rather than a real excess. In population terms, for closely spaced multi-planet radial-velocity systems the high-mass tail implied by unconstrained orientation is largely removed: the surviving configurations cluster near the edge-on masses, so the true masses of these planets are typically close to their tabulated minimum values.

5.3 Parameter-consistency flags

Three systems fail the edge-on consistency test in more than half their realizations: GJ 667 C, HIP 57274, and Kepler-20. These are statements about the published parameters rather than mass limits: the tabulated parameters, with their quoted uncertainties, do not admit a long-lived configuration at any inclination. For GJ 667 C this is consistent with the long-standing debate over the number of planets in the system; for the other two it points to eccentricity solutions that are incompatible with their own dynamics. In each case the appropriate next step is a joint dynamical and radial-velocity refit.

6 DISCUSSION

Three limitations bound the interpretation of the catalogue. First, the inclination sweep is coplanar: each system is tilted as a rigid body, so mutual inclinations enter only through the sensitivity tests of Section 3.2, which shift the limits by at most 0.4 in the mass factor for dispersions up to 5° . Strongly non-coplanar systems, including those the von Zeipel–Lidov–Kozai literature targets, fall outside our assumptions, and their limits should be read as conditional on approximate coplanarity. Secondly, 10^6 inner orbits is short compared with the Gyr ages of these systems. Every depth comparison we have performed, across three generations of the survey, has moved the limits monotonically towards tighter values or left them unchanged, so the reported ceilings are conservative; we make no claim that a surviving configuration is stable

indefinitely. Thirdly, the inputs are catalog parameters sampled within their symmetrized uncertainties, not fits to the radial velocities themselves. The parameter-consistency flags identify where this matters most; for the flagged systems the correct next step is a joint refit rather than a mass limit.

The excluded resonant systems, with adjacent period ratios within a few per cent of a small-integer commensurability, can be treated with the same machinery validated on GJ 876, using single-reference parameters with their fitted phases initialized at the observed libration. Extending the survey to this group would add some of the most closely packed, and therefore most strongly constrained, systems in the Archive.

The catalogue has several direct applications. Planets whose dynamical ceiling lies close to their minimum mass have the narrowest permitted mass range and are the best targets for astrometric confirmation with Gaia. Planets whose ceiling falls below the deuterium-burning limit are established as planetary-mass without additional observations, which removes brown-dwarf contamination from the giant-planet mass function. The parameter-consistency flags identify systems whose published solutions are not supported by their own long-term dynamics and are candidates for reanalysis.

ACKNOWLEDGEMENTS

This work made use of the REBOUND *N*-body package (Rein & Liu 2012) with the WHFast integrator (Rein & Tamayo 2015), and of NUMPY, PANDAS, and MATPLOTLIB. It made use of the NASA Exoplanet Archive, operated by the California Institute of Technology under contract with the National Aeronautics and Space Administration. Claude (Anthropic) was used to assist with code development and manuscript preparation; all results were independently verified by the author, who is responsible for their content.

DATA AVAILABILITY

The code and per-run records are available at <https://github.com/Shawn2highh/dynamass>. The catalogues and per-run records will be archived at Zenodo [TODO: mint DOI at submission]. This work used the NASA Exoplanet Archive `pscompvars` (DOI: 10.26133/NEA13) and `ps` (DOI: 10.26133/NEA12) tables, accessed 2026 July 9–13.

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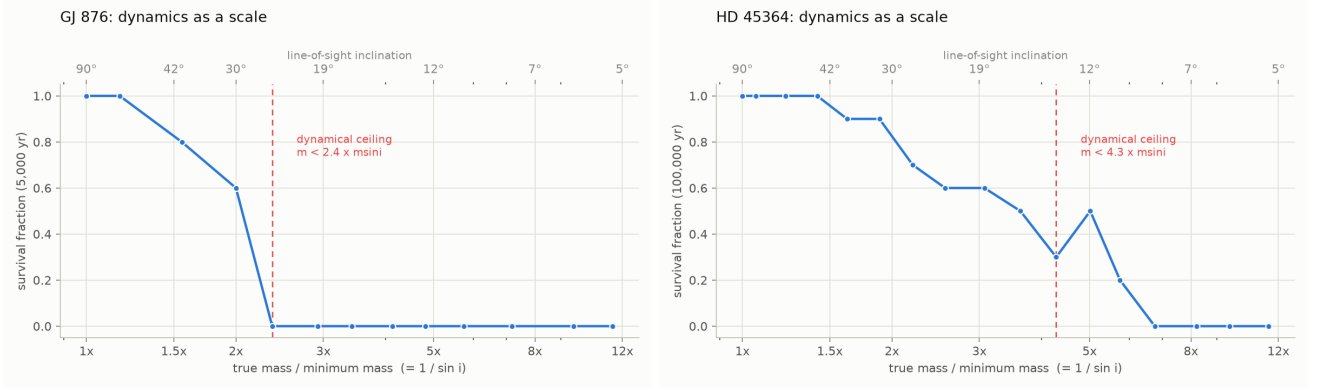


Figure 1. Validation anchors. Left: GJ 876 survival fraction versus inclination with the [Rivera et al. \(2010\)](#) fit phases held fixed; survival is zero at $i \leq 25^\circ$, recovering the literature constraint. Right: HD 45364 ([Li et al. 2022](#)), a smooth, phase-agnostic survival curve (complete survival at $i \geq 45^\circ$, zero by $i \approx 8.5^\circ$).

Table 1. System dispositions (excerpt). Disposition of all 148 qualifying systems: 53 surveyed, 3 flagged (edge-on survival below 0.5), and 92 excluded. The full machine-readable table is available as `table1_dispositions.csv`.

System	Disposition	f_{crit}	$S(90^\circ)$	m_{95}
Barnard's star	surveyed	1.60	0.55	2.18
47 UMa	surveyed	1.34	0.55	2.35
HD 141399	surveyed	3.05	0.85	2.71
K2-18	surveyed	17.92	1.00	2.87
GJ 667 C	flagged	5.71	0.15	—
HIP 57274	flagged	1.23	0.45	—
Kepler-20	flagged	4.25	0.40	—
24 Sex	excluded ^a	1.00	—	—
TOI-500	excluded ^b	32.86	—	—

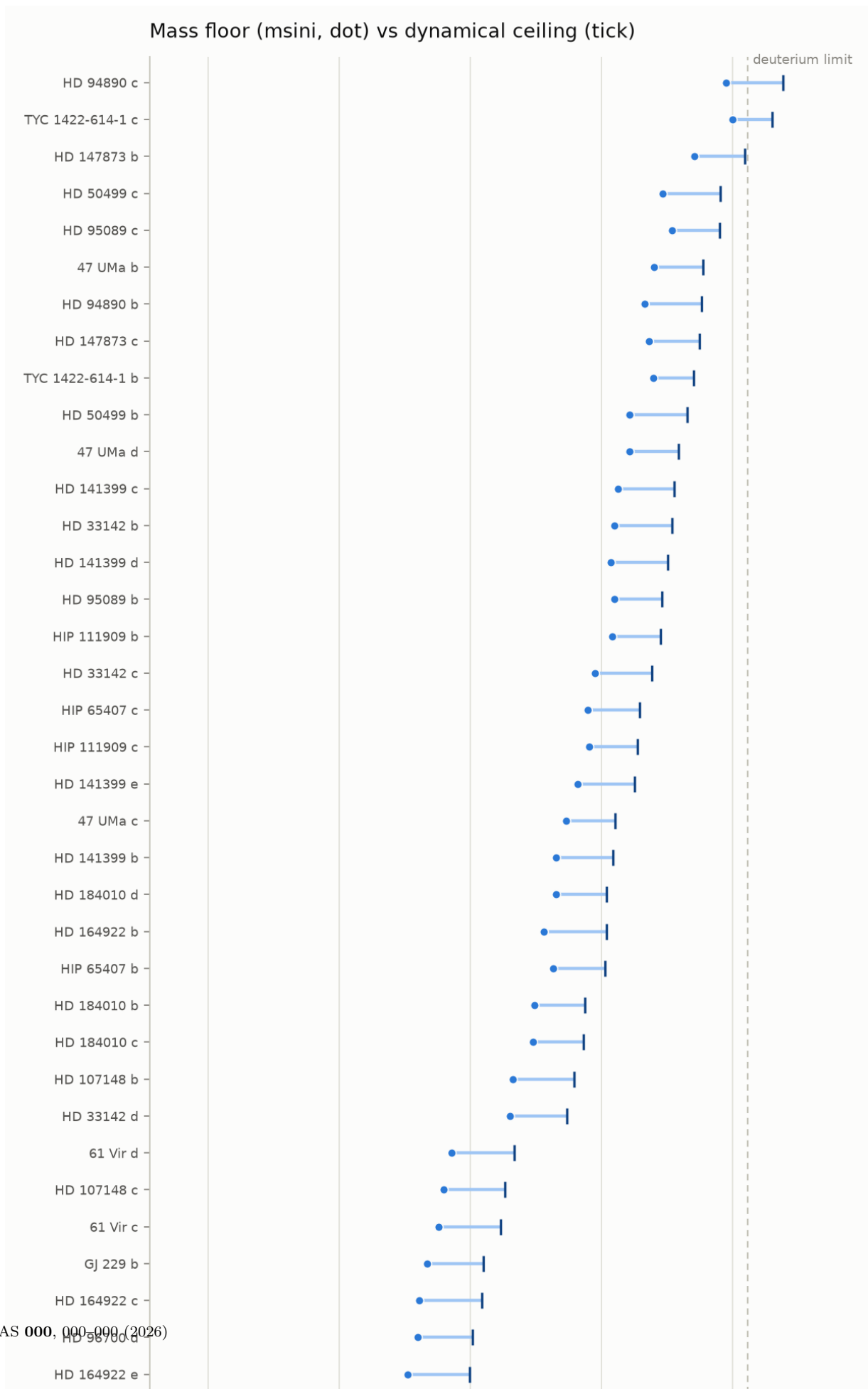
^aResonance-protected / at the stability limit ($f_{\text{crit}} \approx 1$). ^bInner period below 1.5 d.

Table 2. Per-planet 95 per cent credible true-mass limits (ten tightest of 123, sorted by mass factor). The full machine-readable table (`table2_planet_limits.csv`) additionally lists masses in M_{Jup} , the threshold ceiling, and the inclination floor.

System	Planet	$m \sin i (M_\oplus)$	m_{95}	$m_{95} (M_\oplus)$
TYC 1422-614-1	b	794.5	2.01	1600
TYC 1422-614-1	c	3178	2.01	6400
Barnard's star	b	0.299	2.18	0.65
Barnard's star	c	0.335	2.18	0.73
Barnard's star	d	0.263	2.18	0.57
Barnard's star	e	0.193	2.18	0.42
HD 95089	b	400.4	2.31	923
HD 95089	c	1096	2.31	2527
HIP 111909	b	384.5	2.33	897
HIP 111909	c	257.4	2.33	601

Rivera E. J., Laughlin G., Butler R. P., et al., 2010, *ApJ*, 719, 890
 Tamayo D., Gilbertson C., Foreman-Mackey D., 2021, *MNRAS*, 501, 4798
 Vogt S. S., et al., 2014, *ApJ*, 787, 97
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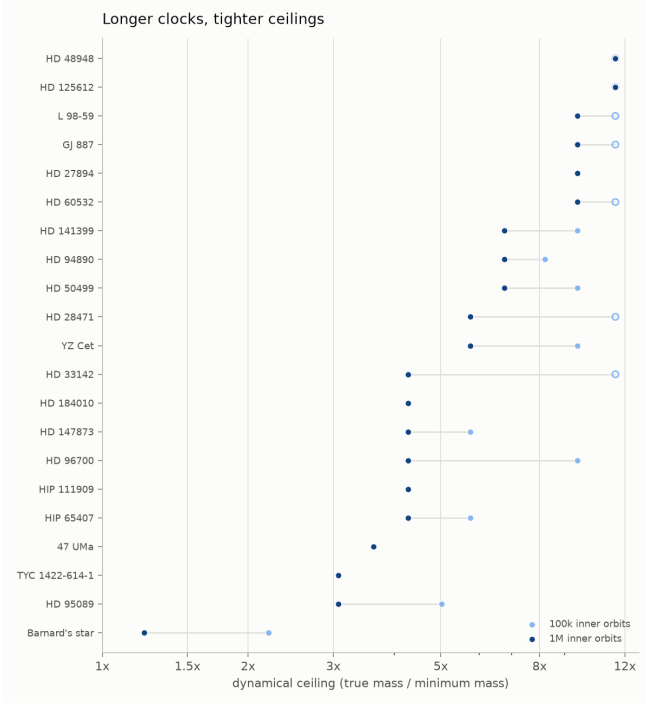


Figure 3. Depth convergence: per-system threshold ceilings at 10^5 versus 10^6 inner-planet orbits with identical parameter draws. Ceilings tighten (9 systems) or are unchanged (5); 7 systems acquire a ceiling only at the greater depth; none loosen.

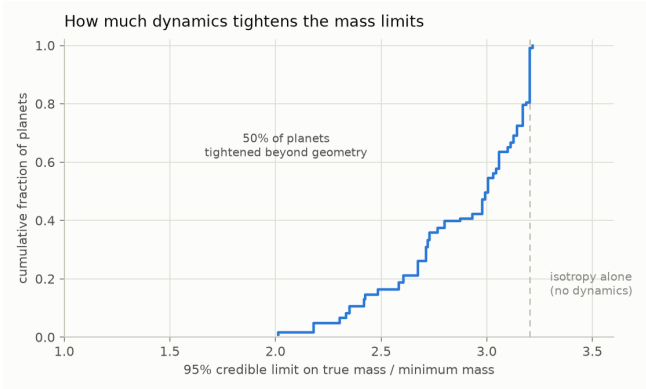


Figure 4. Cumulative distribution of the 123 95 per cent credible mass-factor limits. The dashed line marks the isotropic-prior value (3.20) that geometry alone guarantees; half the sample is constrained inside it.